Hominem Possessions: Full Migration Plan

Date: 2026-10-12 (second pass, evening) · **Status:** FINAL — ready for agent handoff · **Author:** Wyatt (analysis of live Hominem + live sheet; all counts re-verified 2026-10-12 19:25 BST)

> **For the implementing agent — read first.**
>
> - Hominem MCP endpoint: `https://api.ponti.io/api/mcp` (bearer auth). 118 tools total as of this pass; the 13 possessions/container tools in §2 are unchanged — nothing in §6 is built yet.
> - Source spreadsheet: `possessions`, id `1j-0hTJZIoPdycWB-gHa1DKCa7arMLwiL2ygJTn1uox0`. It now has exactly 4 tabs: `master`, `containers`, `luggage`, `weights` (all view tabs were deleted 2026-10-12; see §3).
> - Ground rules: map sheet columns **by position, never by header name** (§3). Never use Sheets `values.append` on this file (§3 failure modes). Preserve `notes` verbatim. When this doc and live data disagree, live data wins — re-run the diffs in §9 Phase 2 before importing.
> - Do not build anything in §6's "Explicitly not building" list.

1. Objective

Make Hominem the single source of truth for Charles's possessions, replacing the Google Sheet as the working registry. The sheet becomes a frozen backup. All of my inventory work (move planning, "do I own X", weight rollups, audits) moves to Hominem tools.

Guiding principle: Hominem should model what the sheet *means*, not what it *looks like*. The sheet's complexity is mostly accidental (views-as-tabs, position-based columns, formula rollups). The plan eliminates the accidental complexity and keeps the essential: items, containers, nesting, weights, money, lifecycle.

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2. Current Hominem capabilities (audited live 2026-10-12)

Tools that exist — 13 total, all working

Tool	Mode	Notes
`possession_create`	write	20 fields; `name` required. **No `externalId` param.**

Tool	Mode	Notes
`possession_update`	write	`{id, data}` — partial update, metadata merged. **No `externalId`, no `weightKg`, no `moveAction` in `data`.**
`possession_get`	read	By id
`possession_list`	read	Filters: status, archived, containerId, category, query. Max 100/page, offset-based
`possession_move`	write	Batch ≤50 into/out of a container
`possession_delete`	write	Hard delete (guidance prefers archive/dispose)
`possession_summary`	read	Counts by status/category, value by currency (20 currencies)
`container_create` / `container_update` / `container_get` / `container_list` / `container_move` / `container_delete`	mixed	Nesting via `parentContainerId`, cycle rejection, `containerKind: durable_possession` for item-that-is-a-container (e.g. Gucci boxes)

Data already in Hominem (verified, deduped by id)

- **428 possessions:** 284 active + 144 archived. No duplicate externalIds (verified — an early count of "144 dups" was a pagination artifact, see §8).
- Active = 107 `SHOP-xxx` (shopping-sheet import) + 177 `ITM-xxx` (possessions-sheet import, Sep 30).
- Archived = 144 `ITM-xxx`, status `disposed` — the sheet's non-owned rows, correctly imported as archived.
- **40 containers:** 39 with `CON-xxx` externalIds + 1 Hominem-native ("video game pouch", no externalId). Sheet↔Hominem container drift is itemised in Phase 2.
- Import metadata is clean: `metadata.importedFrom`, `sourceItemId` on containers.

What's missing (gaps)

1. **No `weightKg` / `volumeCbm` on possessions.** Containers have them; items don't. The entire weights-tab workflow (vessel packing limits) cannot be reproduced.
2. **No `externalId` on create/update.** The Sep 30 import must have used a backdoor. Without it, no idempotent re-import or sync is possible via public tools.

3. **No `moveAction`.** The sheet's vessel assignment (`luggage_medium`, `send my bag box 1`) has no field. `containerId` covers items *in* containers, but move_action is the coarser "which vessel/box" routing used by the weights tab.
4. **Status enum lacks `donated`, `sold`.** The sheet uses both (7 donated, 2 sold in master). Currently unmappable except via metadata.
5. **The 4 new master columns have no home:** `acquisition_type`, `purchase_currency`, `retirement_method`, `ownership_type` → metadata at best.
6. **No weight rollup.** Nothing computes "contents kg per vessel" — the weights tab's core function.
7. **No history/audit.** The sheet has no changelog either, but a real system should.
8. **Unstable pagination.** `possession_list` sorts by `createdAt`; all imported rows share one timestamp, so offset pages repeat/skip rows. (This bit me during this analysis.)

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3. Spreadsheet complexity analysis (live read 2026-10-12)

Spreadsheet: `possessions` (`1j-0hTJZIoPdycWB-gHa1DKCa7arMLwiL2ygJTn1uox0`).

Current tabs (4 — analysis found only ONE item registry)

Tab	Rows	Cols	What it is
`master`	352	24	**The item registry.** All unique items live here.
`containers`	38	12	Container registry (CON-xxx)
`luggage`	3	—	Packing list (different shape, not item rows)
`weights`	13	8	Vessel weight rollup (formulas)

Unique items in the sheet: **352** (master).

Removed during this analysis (2026-10-12, verified lossless): six view tabs — `jewelry` (37 rows), `art` (52), `archive` (146) were stale static copies of master subsets (100% ID overlap; 27 drifted cells, every one data the view had lost, zero unique values); `selling` and `clothing` were broken (`#REF!` in A1, no data); `tattoos` was an empty header-only view. Nothing referenced them (post-deletion scan: zero `#REF!` in remaining tabs). Google Sheets version history can restore them, but there is no reason to — in Hominem they are `possession_list` filters.

Schema (master, by position — headers are unreliable)

24 columns: item_id(A) · name(B) · brand · status · category · sub_category · model · acquired_date · retired_date · purchase_price · sale_price · serial_number · size · color · container_id(O) · url ·

move_action(Q) · notes(R) · weight_kg(S) · vol_cbm(T) · acquisition_type(U) · purchase_currency(V) · retirement_method(W) · ownership_type(X).

- **Headers are renamed live** (Oct 2: A="Column 1" actually holds item_id). All automation must map by position, never by header name.
- **Statuses in use:** owned 205 · disposed 137 · donated 7 · sold 2.
- **Newer columns** (values observed): `acquisition_type` ∈ {purchased, unknown, created_from_existing_item, gift_or_inherited}; `purchase_currency` = USD wherever filled; `retirement_method` ∈ {discarded, donated, gifted, sold}; `ownership_type` ∈ {functional, wearable_identity, personal_archive, storage_transport, collectible}.
- **Dual container semantics** (the subtlest part): col O holds the *innermost* container (CON-xxx); col Q (`move_action`) holds the *vessel* (suitcase name or `send my bag box N`). Camera gear sets Q = container name; clothing sets Q = vessel. The weights tab sums **both** conventions per vessel — any replacement must replicate this exactly.

Weights tab (the logic to replace)

13 rows: per vessel — contents kg (Σ item weights by vessel, both conventions) · case tare kg · total · airline limit kg · notes · ✓/flag. Plus rollup rows: cabin total 6.90/12kg, checked total 14.73/46kg. Includes judgment calls in Notes (e.g. "ITM-0033 tags 'vacuum_shirts_tees' (no such bag) — counted here"). A pure SUMIF replacement loses these annotations — they need a home (vessel notes field).

Failure modes observed

- **Broken derived views** (`selling` , `clothing` → `#REF!` , root cause unconfirmed; presumed dead cross-tab reference). Deleted 2026-10-12 — do not recreate; Hominem queries replace them.
- **Append hazard:** gap rows exist; `values.append` stops at the first empty row and can overwrite live data. (Burned us Sep 15.) Read/write by explicit row resolved from item_id, or don't write to the sheet at all.
- **Live-edit risk:** Charles edits cells between reads; row numbers shift on re-sort (burned Sep 17 — wrote to the wrong row). Always resolve rows by item_id from a fresh read immediately before any write.
- **Orphan container refs (verified 2026-10-12):** master col O references two containers with no row in the `containers` tab — **CON-024 (11 items)** and **CON-036 (1 item)**. CON-024 *does* exist in Hominem; CON-036 exists in *neither* system. Both are handled in Phase 2; `possession_validate` must catch this class permanently.

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4. Gap analysis: sheet concept → Hominem home

Sheet concept	Hominem today	Needed
item row (24 cols)	`possession_*` (20 fields)	+ `weightKg`, `volumeCbm`, `moveAction`, `externalId` (create/update), `acquisitionType`, `purchaseCurrency`, `retirementMethod`, `ownershipType`
statuses donated/sold	enum lacks them	extend enum (or map + metadata — recommend extend)
containers tab	`container_*` complete	+ `trackingReference` field (sheet col L has it; Hominem doesn't)
jewelry/art/archive/tattoo views	—	**nothing — DONE.** Tabs deleted 2026-10-12; replaced by `possession_list` filters (category/status/archived).
selling/clothing views	—	**nothing — DONE.** Tabs deleted 2026-10-12 (were broken anyway); replaced by `status=wishlist` / category queries.
weights tab	—	`vessel_weight_report` (computed) + vessel notes
luggage packing list	—	a `packing_list` note or container `description`; not a data model
live header renames	n/a	Hominem fields are stable by definition — this whole failure class disappears
no changelog	—	`possession_history` (new)

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5. Target data model (additions to what exists)

Possession — add optional fields: `weightKg` (number ≥0), `volumeCbm` (number ≥0), `moveAction` (string ≤100, the vessel token), `externalId` (string, unique, settable on create), `acquisitionType` (enum), `purchaseCurrency` (3-letter; separate from display `currencyCode` if they differ), `retirementMethod` (enum), `ownershipType` (enum). Status enum += `donated`, `sold`.

Container — add `trackingReference` (string ≤200, for the Send My Bag refs).

Vessel — not a new entity. A vessel is either a container (box) or a possession-with-`containerKind=durable_possession` (suitcase). `moveAction` on items points at the vessel by name token; the weight report resolves tokens → vessels. Keep the sheet's dual convention, but make it

explicit and validated (unknown token → validation warning, not silent mis-sum — this is what caught the `vacuum_shirts_tees` ghost).

PossessionEvent (new, append-only): `{possessionId, at, actor, kind, diff}` — kind ∈ {created, updated, moved, status_changed, archived, deleted}. Powers `possession_history`.

5.1 Column mapping — master (by position) → Possession

Pos	Sheet column	→ Hominem field	Transform
A	item_id	`externalId`	verbatim (`ITM-xxx`); the upsert key
B	name	`name`	verbatim
C	brand	`brand`	empty → null
D	status	`status` (+ `isArchived`)	`owned` → owned; `disposed` / `donated` / `sold` → same-named status **and** `isArchived=true`
E	category	`category`	verbatim (lowercase tokens)
F	sub_category	`subCategory`	empty → null
G	model	`model`	empty → null
H	acquired_date	`acquiredDate`	parse to ISO date; unparseable → null + flag in import report
I	retired_date	`retiredDate`	same
J	purchase_price	`priceCents`	sheet stores formatted strings (`£30.00`, `$1,234.56`): strip symbol/commas → minor units; blank → null
K	sale_price	`sellPriceCents`	same
L	serial_number	`serialNumber`	**always string** (leading zeros exist)
M	size	`size`	verbatim free text (dimensions, `King`, shoe sizes)
N	color	`color`	empty → null
O	container_id	`containerId`	`CON-xxx` → resolve via container `externalId`; unresolvable (CON-036) → null + flag

Pos	Sheet column	→ Hominem field	Transform
P	url	`url`	empty → null
Q	move_action	`moveAction` *(new)*	verbatim token (`luggage_medium` , `send my bag box 1` , container names)
R	notes	`notes`	**verbatim** — contains judgment calls; never summarise
S	weight_kg	`weightKg` *(new)*	numeric string → number; blank → null
T	vol_cbm	`volumeCbm` *(new)*	same
U	acquisition_type	`acquisitionType` *(new)*	enum §3; blank → null
V	purchase_currency	`purchaseCurrency` *(new)*	ISO 3-letter; blank → infer from price symbol (J), else null + flag. (Filled on only 176/352 rows, all USD.)
W	retirement_method	`retirementMethod` *(new)*	enum §3; blank → null
X	ownership_type	`ownershipType` *(new)*	enum §3; blank → null

Containers tab → Container: A `container_id` → `externalId` (upsert key) · B `name` · C `container_type` → `containerType` · D `status` · E `container_kind` → `containerKind` · F `source_item_id` → `metadata.sourceItemId` · G `parent_container_id` → resolve via externalId → `parentContainerId` · H `weight_kg` → `weightKg` · I `vol_cbm` → `volumeCbm` · J `notes` → `description` · K `current_location` → `currentLocation` · L `tracking_reference` → `trackingReference` *(new)*.

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6. Tool inventory

Keep as-is (13)

`possession_create/get/list/update/delete/move/summary` , `container_create/get/list/update/move/delete` . Minor fix: stable pagination on list tools (sort by `createdAt, id` , or cursor-based).

New tools needed (10)

1. **`possession_upsert`** — create-or-update keyed on `externalId`. Accepts the full field set incl. `externalId`. Returns `{id, created: bool}`. *The* migration primitive; also the future sync primitive. Mode: write.
2. **`possession_bulk_upsert`** — up to 100 items per call, per-item `{externalId, id, created, error}` results, atomic=false (partial success reported). For the initial load and any re-import.
3. **`container_upsert`** — same pattern keyed on `externalId` (CON-xxx).
4. **`possession_history`** — `{id, limit}` → append-only event list. Mode: read-only.
5. **`vessel_weight_report`** — computed packing report replacing the weights tab: per vessel `{vessel, contentsKg, tareKg, totalKg, limitKg, notes, overLimit: bool}` + cabin/checked rollups. Vessel set = distinct `moveAction` tokens + containers with kind vessel. `limitKg` / `tareKg` / `notes` stored on the container (new optional fields) or a small vessel-config; item weights summed across the subtree honoring both placement conventions. Mode: read-only.
6. **`possession_validate`** — integrity audit replacing my manual sheet audits: unplaced active items, items missing weight, `moveAction` tokens with no vessel, orphan `container_id` refs (the CON-036 class), negative/zero prices, `disposed` items not archived, archived items still in a container. Returns findings list. Mode: read-only.
7. **`possession_search_advanced`** *(optional)* — the existing `query` matches name/brand/model; extend to notes + serial number, or rely on `semantic_search`. Decide during build; don't duplicate.
8. **`container_contents_weight`** — folded into #5; listed here only to say *don't* build it separately.
9. **`move_plan`** *(phase 2, optional)* — a named packing plan: `{name, vesselAssignments: {possessionId: moveAction}}` applied via `possession_move` + `moveAction` update in one call, dry-run first. The sheet's move planning is currently manual cell edits; this is the long-term payoff feature, not migration-critical.
10. **`possession_merge`** — *not needed.* Verified zero duplicate externalIds. Do not build speculatively.

Explicitly not building

- Anything replicating the jewelry/art/archive/tattoo tabs (they're saved filters).
- Finance-link tools (value rollups already in `possession_summary`; the finance sheet stays canonical).
- Calendar/email/Tasks mirrors.

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7. REST API endpoints

Hominem is an HTTP API with MCP as one surface; every tool maps to an endpoint. Proposed (under the existing `/api` prefix):

```
GET    /possessions                    # list (filter/sort/paginate; stable sort)
POST   /possessions                    # create (accepts externalId)
POST   /possessions/upsert             # keyed on externalId
POST   /possessions/bulk-upsert        # ≤100, per-item results
GET    /possessions/:id
PATCH  /possessions/:id
DELETE /possessions/:id                # confirm-gated
POST   /possessions/move               # batch ≤50
GET    /possessions/:id/history
GET    /possessions/summary
GET    /possessions/validate           # integrity audit
GET    /containers
POST   /containers
POST   /containers/upsert
GET    /containers/:id                # incl. children + possessions + counts
PATCH  /containers/:id
DELETE /containers/:id
POST   /containers/:id/move           # re-parent
GET    /reports/vessel-weights         # packing report
```

Conventions: UUID primary keys, `externalId` unique secondary key (ITM-xxx / CON-xxx / SHOP-xxx namespaces), prices integer minor units + ISO currency, dates ISO-8601, soft-delete via `isArchived` (hard DELETE confirm-gated), every write emits a PossessionEvent/ContainerEvent.

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8. Verification notes (how the analysis was done — and a bug caught)

- All counts verified by deduping on record `id`: **428 unique possessions** (284 active, 144 archived), 40 containers.
- **Pagination trap:** `possession_list` offset pages repeat rows when `createdAt` ties (all Sep-30 imports share one timestamp). Naive page-and-concatenate over-counted 572 and phantom "duplicates." Fix in §6 (stable sort); analysis above uses deduped figures.
- **Exact sheet ↔ Hominem diff (re-verified 2026-10-12 19:25):** missing from Hominem = **ITM-0409 → ITM-0440** (32 items, contiguous — everything added after the Sep 30 import) + **CON-041** (1 container). Hominem-only = **ITM-0400**, **CON-040**, **CON-024** (see Phase 2). No other drift.
- **The deleted view tabs were mirrors, not backlog:** all their IDs existed in master. The migration backlog is exactly the diff above — nothing else.

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9. Migration phases

Phase 0 — Decide & freeze protocol. Charles confirms Hominem becomes canonical. Agree a cutover window (he edits the sheet live; a freeze or a final-diff pass is required — recommend: he stops editing for one evening, or we do a final `bulk-upsert` diff the morning after).

Phase 1 — Build the gaps (§5–§6). Fields (`weightKg` , `volumeCbm` , `moveAction` , `externalId` on create/update, 4 enum columns, `trackingReference` , status enum += donated/sold), `possession_upsert` / `bulk-upsert` / `container_upsert` , `possession_history` (+ event emission on all writes), stable pagination, `vessel_weight_report` , `possession_validate` .

Phase 2 — Backlog import. Re-run both diffs first (live data wins over this list), then import via `bulk-upsert` keyed on `externalId` , using the §5.1 mapping. Verified backlog at 2026-10-12 19:25:

- **Import 32 items:** `ITM-0409` → `ITM-0440` (contiguous; everything added to master after the Sep 30 import).
- **Import 1 container:** `CON-041` (in the sheet registry, not in Hominem).
- **Reconcile Hominem-only records (do not delete without Charles's word):** `ITM-0400` and `CON-040` exist in Hominem but not the sheet (added directly, or sheet rows since deleted). Default: keep, flag in the import report.
- **CON-024:** referenced by 11 master items, missing from the sheet's container registry, present in Hominem. Treat the Hominem record as authoritative; item references resolve against it. No sheet fix needed (the sheet is being retired).
- **CON-036:** referenced by 1 master item, exists in neither system. Import that item with `containerId=null` and flag it in the report + `possession_validate` output. Do not invent a container record.
- **"video game pouch"** (Hominem container, no externalId): assign the next free `CON-xxx` externalId during this phase so every container is addressable by the upsert key.
- Transform rules: §5.1. Prices → minor units; empty strings → null; `disposed/donated/sold` rows → matching status + `isArchived=true` ; `notes` verbatim.

Phase 3 — Verification. Done when ALL of these pass:

- Counts: every master item_id and containers-tab id present in Hominem exactly once by externalId (352 + post-import additions; 39 CONs + reconciled records). A re-run of `bulk-upsert` with the same input creates 0 records (idempotency check).
- `vessel_weight_report` output vs weights tab, row by row (tolerate rounding; investigate any >0.05kg diff — the ghost-bag class of bug).
- `possession_summary` value totals vs finance sheet expectations.
- `possession_validate` output contains no un-triaged findings (CON-036 flag is expected and pre-triaged).
- Spot-check 20 high-value items field-by-field against master (incl. one `created_from_existing_item` , one with a serial number with leading zeros, one in CON-024).

Phase 4 — Replace the views. ✅ **DONE 2026-10-12.** All six view tabs deleted (selling, clothing, tattoos, jewelry, art, archive), verified lossless (zero unique data; all drift was view-side loss; zero `#REF!` in the remaining tabs). Their replacements are `possession_list` filters — no filter definitions needed preserving beyond that. Only remaining item: decide whether the `luggage` packing-list content (3 rows) becomes a note/ `packing_list` somewhere in Hominem — trivial, fold into Phase 5.

Phase 5 — Cutover. Sheet → read-only (or a dated frozen copy; a pre-consolidation backup already exists from Sep 14). Update my operating notes: retire the column-position map, point all inventory workflows at Hominem tools. Keep the sheet file as the frozen backup, not a living doc.

Phase 6 — Long-term ops.

- Schema evolution: new attributes → optional fields or `metadata`, never column renames; MCP is versioned — additive changes only.
- The 4 enum columns will grow new values (Charles invents categories live) — enums must be extendable without a deploy, or modelled as free strings with suggestions.
- `possession_validate` on a schedule (weekly) to catch orphan refs and missing weights.
- `move_plan` (tool #9) when the next move approaches — this is where Hominem starts paying back more than the sheet ever did.

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10. Open questions for Charles

1. `donated` / `sold` as first-class statuses, or map to `disposed` + `retirementMethod`? (Recommend: first-class — the sheet already treats them as statuses.)
2. Should the 4 new columns (`acquisition_type`, `purchase_currency`, `retirement_method`, `ownership_type`) be first-class fields or `metadata`? (Recommend: first-class — they're load-bearing now.)
3. Cutover style: hard freeze evening, or rolling dual-write with a final diff?
4. Keep the frozen sheet in Drive as the backup, or export to CSV in the repo too?
5. Vessel limits/tares/notes (weights-tab config): store on the container records, or a separate small config?
6. The `SHOP-xxx` possessions (107, from the shopping sheet) — are they inventory or a separate concern? They currently share the registry.
7. Hominem-only records `ITM-0400` and `CON-040` (not in the sheet) and the externalId-less "video game pouch" container: keep as Hominem-native (recommended), or treat the sheet as truth and remove/align them?